

Liquid Assets: Blood Plasma Donation Income and Crime

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Motivation

- Crime is costly: \$2.6 to \$5.8 trillion annually in the U.S.
 - Traditional responses (policing, incarceration) are expensive and can backfire.
- Economic interventions outside the criminal justice system can unintentionally reduce crime.
 - Examples: lead remediation [[Reyes, 2007](#); [Aizer and Currie, 2019](#)], educational expansion [[Lochner and Moretti, 2004](#); [Fella and Gallipoli, 2014](#); [Bell et al., 2022](#)]
- This paper: tests whether access to compensated plasma donation—a **flexibly timed, conditional income source**—can reduce crime.
- Research Questions
 - **RQ-1:** Does the opening of a plasma center impact local crime?
 - **RQ-2:** Does plasma-center access lower recidivism among formerly incarcerated individuals?
 - **RQ-3:** Through what mechanisms do effects arise?

Contributions of the Paper and Preview of Results

- Extends research on financial incentives and crime by studying a setting where income is:
 - **flexibly timed** (up to 2×/week), allowing individuals to smooth short-term financial strain
 - **conditional**, requiring compliance with drug-screening and eligibility rules
- Provides the first evidence on how flexibly timed, conditional income from plasma donation centers affects crime
 1. **City-level**: difference-in-differences design using NIBRS $\Rightarrow$ large reductions in total, property, and drug-related crime, especially pronounced in poorer cities
 2. **Neighborhood-level**: within-city design using distance-based design $\Rightarrow$ localized total and property crime drops near new centers
 3. **Individual-level**: reentry-based analysis using NCRP $\Rightarrow$ reduced recidivism, especially for younger and economically disadvantaged individuals

Blood Plasma Donation: Market Scale and Donor Demographics

- Plasma an essential input for life-saving therapies
 - Immunodeficiency, COVID, bleeding disorders, etc.
 - US supplies $\approx$ 70% of global plasma
- Plasma donations are **compensated** in the U.S.
 - \$30 to \$70 per donation (2x per week). 2019 average = \$58
 - 2019: 53M donations for an estimated \$3 billion compensation [[Dooley and Gallagher, 2024](#)]
→ Estimated 2.2M donors
- Donor demographics & economic constraints [[Dooley and Gallagher, 2024](#)]
 - **Young, median income <\$20K, poor credit, limited savings**

Blood Plasma Donation: Step-by-Step Overview

1. Registration & Screening (~30 mins)

- **Questionnaire:** Donor self-reports medical history, drug use, incarceration, and lifestyle risks
 - Temporary deferrals (self-reported, unverifiable): jail $\geq$ 3+ (past 12mo), injection drug use (past 3mo)
 - Permanent deferrals (lab-verified, national blacklist): HIV or hepatitis
- **Medical Check:** Nurse assesses vitals, medical history, and visual signs of drug use or impairment
 - FDA criteria: age 18-64, $\geq$ 110 lbs, no disqualifying conditions [▶ All Conditions](#)
- **Laboratory Testing:** Plasma screened for infectious diseases (HIV, hepatitis, syphilis)

2. Apheresis Procedure & Recovery (~60-90 mins)

- Blood is drawn and passed through a machine that separates plasma from other components
- Plasma replenishes naturally within 48 hours [▶ Side Effects](#)
- Donation frequency: 2 donations per 7 days

3. Compensation (via prepaid debit or ATM cards)

- **Monetary compensation is between \$30 and \$70 per session** (2019 average ~\$58)

Data Sources

1. Plasma Center Data (2000-2023) [[Dooley and Gallagher, 2024](#)]
 - Plasma center **location** and **opening date**
2. National Incident Based Reporting System (NIBRS) (2005-2019)
 - “Nationwide” incident-level database of reported crimes
 - **City, time, crime type** and **offender characteristics**
 - Data voluntarily provided by police agencies to FBI
3. Crime Open Database (CODE) [[Ashby, 2018](#)]
 - Database of reported crimes across 21 cities
 - Offense **crime type, date** and **location**
4. National Corrections Reporting Program (NCRP) (2000-2019)
 - Individual-level administrative data on prison admissions and releases
 - Includes **offense type, demographics, and prior criminal history**

▸ data map

▸ plasma centers location characteristics

Design 1: City-Level Crime Outcomes

$$C_{cst} = \beta D_{ct} + \gamma_c + \theta_{s \times t} + \epsilon_{cst}, \quad (1)$$

- D_{ct} takes value 1 if at city c at period t there is a blood plasma center, and 0 otherwise.
- C_{cst} crime rate in city c in state s in period t
- γ_c city fixed effects
- $\theta_{s \times t}$ state-by-year-by-month fixed effects
- ϵ_{cst} clustered at the city level
- Identification: PTA 
- Is TWFE design valid here? 
 - Goodman-Bacon [2021] decomp: **clean comparisons comprise 98% TWFE weight**

[ES Graphs]

[Alternative Estimators]

The Impact of Plasma Centers on City-Level Crime Rates

► ES

► DR

► prop

► drugs

	(1)	(2)	(3)	(4)	(5)	(6)
	Total Crime	Violent Crime	Property Crime	Drug Sales	Drug Possession	Other Crime
Plasma Center in City	-76.3*** (21.3)	-9.11* (5.29)	-48.3*** (15.5)	-.567 (1.16)	-17** (7.09)	-1.25** (.485)
Proportion of Total Crime	[1.000]	[0.197]	[0.584]	[0.022]	[0.182]	[0.015]
$\bar{Y}_{NT}$	648	127	377	14.7	119	9.34
Plasma Center/ $\bar{Y}_{NT}$	-.118*** (.0329)	-.0715* (.0415)	-.128*** (.0411)	-.0386 (.0791)	-.143** (.0594)	-.134** (.0519)
Number Treated Cities	53	53	53	53	53	53
Number Cities	1,185	1,185	1,185	1,185	1,185	1,185
Observations	198,744	198,744	198,744	198,744	198,744	198,744
Adjusted R^2	.718	.678	.707	.274	.498	.247

Additional City-Level Crime Results

- **Long-run Dynamics:** Suggests building effect over time [\[ES Graphs\]](#)
- **Distribution Regression:** Large impacts at upper end of crime distribution [\[DR\]](#)
- **Types of Property Crimes:** Theft and car theft are key components of property crime drop [\[prop\]](#)
- **Drug Possession by Substance:** Marijuana and Methamphetamine are key components of drop in drugs possession [\[drugs\]](#)
- **Offender age:** Effects decline with age, largest at 18–24. [\[age\]](#)

$$C_{gct} = \beta D_{gt} + \gamma_g + \theta_{c \times t} + \epsilon_{gct}, \quad (2)$$

Identification approach à la [Linden and Rockoff \[2008\]](#) and [Ang \[2021\]](#)

- C_{gct} is the crime count in census block group g in city c in period t .
- D_{gt} takes value 1 for areas within a 3 km radius of a plasma center once it opens, and 0 otherwise. [\[why 3 km?\]](#)
- γ_g block group fixed effects
- $\theta_{c \times t}$ city-by-year-by-month fixed effects
- ϵ_{cst} clustered at the block group-by-plasma center level
- Identification: PTA  

[\[ES Graphs\]](#)[\[Placebo\]](#)

The Impact of the Opening of a Nearby Blood Plasma Center on Block-Group Crime Counts

► ES

► placebo

	(1)	(2)	(3)	(4)	(5)
	Total Crime	Violent Crime	Property Crime	Drug Crime	Other Crime
Near Plasma Center $\times$ Open	-.318** (.153)	-.107** (.054)	-.26** (.105)	.0185 (.0268)	.0309 (.039)
Proportion of Total Crime	[1.000]	[0.238]	[0.567]	[0.062]	[0.133]
$\bar{Y}_{NT,pre}$	11.7	2.76	6.57	.8	1.61
DD/ $\bar{Y}_{NT,pre}$	-.027** (.013)	-.0387** (.0195)	-.0396** (.016)	.0232 (.0335)	.0192 (.0242)
Observations	52,440	52,440	52,440	52,440	52,440
Adjusted R^2	.805	.63	.747	.502	.588

- Effects are smaller in magnitude than city-level results (NIBRS), reflecting short-run focus and dataset coverage.

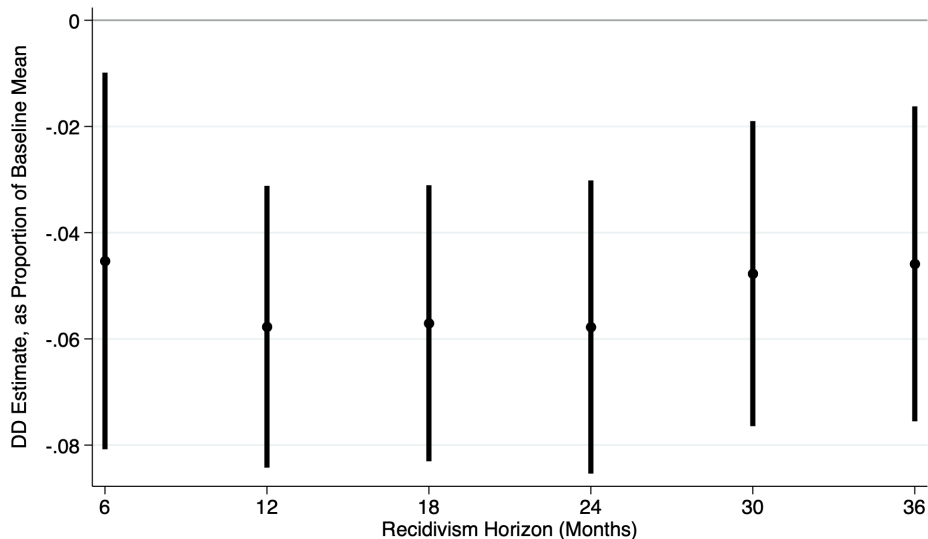
[Reconciling?]

Design 3: Individual-level Recidivism

$$Y_{icst} = \beta D_{cst} + \gamma X_i + \theta_{s \times t} + \kappa_c + \epsilon_{icst} , \quad (3)$$

- Y_{icst} recidivism measure (=1 if prisoner returns to prison within a certain period of time)
- D_{cst} treatment (=1 if an individual is released after a plasma center opens, and 0 otherwise.
- X_i individual characteristics (e.g., age, gender, sentence)
- κ_c county fixed effects and $\theta_{s \times t}$ state-by-year-by-month fixed effects
- ϵ_{icst} clustered at the county level
- Data: National Corrections Reporting Program (NCRP)

The Impact of Plasma Centers on Recidivism Rate

[► ES](#)[► Raw](#)

The Impact of Plasma Centers on 3-year Recidivism Rate by Crime Type

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Overall	Violent	Property	Drug Possession	Drug Sales	Weapons	DUI	Other
Treated	-.0166*** (.00548)	.00255 (.00243)	-.00638* (.00329)	-.00762** (.00363)	-.0115*** (.00300)	-.0000723 (.00124)	.000371 (.00129)	-.00595* (.00347)
$\bar{Y}_{NT}$	.362	.0881	.146	.0550	.0636	.0196	.0196	.0938
Treated/ $\bar{Y}_{NT}$	-.0459*** (.0151)	.0289 (.0276)	-.0437* (.0225)	-.139** (.0660)	-.181*** (.0472)	-.00369 (.0633)	.0189 (.0658)	-.0634* (.0370)
Observations	2,592,042	2,592,042	2,592,042	2,592,042	2,592,042	2,592,042	2,592,042	2,592,042
R^2	.0874	.0414	.0566	.0453	.0583	.0172	.0377	.0826

1.) Evidence in support of an **income channel** driving effects

- Pronounced heterogeneity across ex-ante city-level, income-related dimensions: male unemployment rate, median household income, and poverty rate [income channel: city-level]
- Stronger effects among lower-income terciles: recidivism declines most for individuals in the bottom predicted-income group [income channel: individual-level]

2.) Little support for an **incapacitation effect** of donation driving effects

- DDD analysis [at city-hour-day-month-level] using opening hours of plasma clinics as the third difference [incapacitation channel]

Conclusions and Policy Implications

- Plasma centers create accessible, conditional income streams for donors.
- Opening of a plasma center leads to drop in crime, notably property and drugs crime
 - Mechanism 1: Income channel 
 - Mechanism 2: Incapacitation channel 
- Policy implications:
 - Plasma income (up to $\sim$ \$400/500 per month) parallels small-scale UBI experiments.
 - Prior evidence shows mixed effects of unconditional transfers on crime and substance use.
 - In this setting: \$400/500 per month + flexibly timed + conditionality (drug-screening) $\Rightarrow$ declines in property and drug-related crime.
 - even the case for recidivism – notoriously difficult to reduce
 - Highlights the importance of **conditionality and timing** in cash-transfer design.

References

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- Desmond Ang. The effects of police violence on inner-city students. *The Quarterly Journal of Economics*, 136(1):115–168, 09 2021.
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- Jeffrey M Wooldridge. Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators. *Available at SSRN 3906345*, 2021.

- Eligibility Criteria
 - Age 18-64, $\geq$ 50 lbs
 - In “good” health on day of donation
 - Valid government ID and permanent address
- Temporary Deferral
 - Incarceration spell of 3+ days in the past 12 months
 - Injection drug use in past 3 months
 - Tattoo or piercing in past 4 months
 - Pregnancy
 - Travel to certain regions
 - Health conditions (e.g., anemia, heart disease)
 - Risky sexual behaviors
- Permanent Deferral (National Blacklist)
 - Positive test for HIV or Hepatitis
 - Attempt to donate at multiple clinics

Side Effects

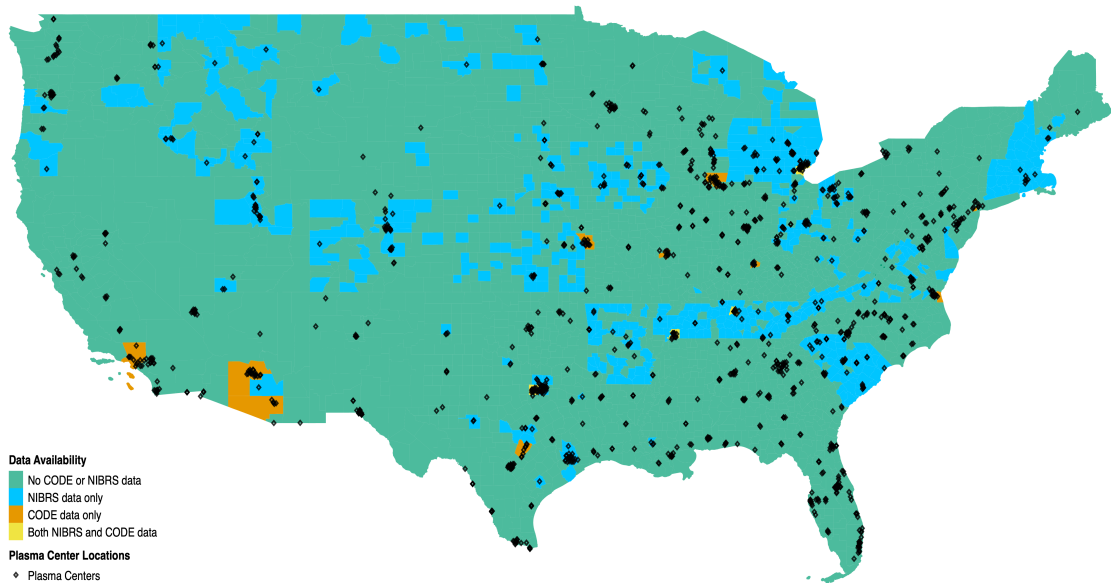
Common side effects

- Fatigue
- Dizziness
- Bruising

Rare Risks

- Fainting
- Infection at needle site

Data Availability and the Location of Plasma Centers

[◀ Back](#)

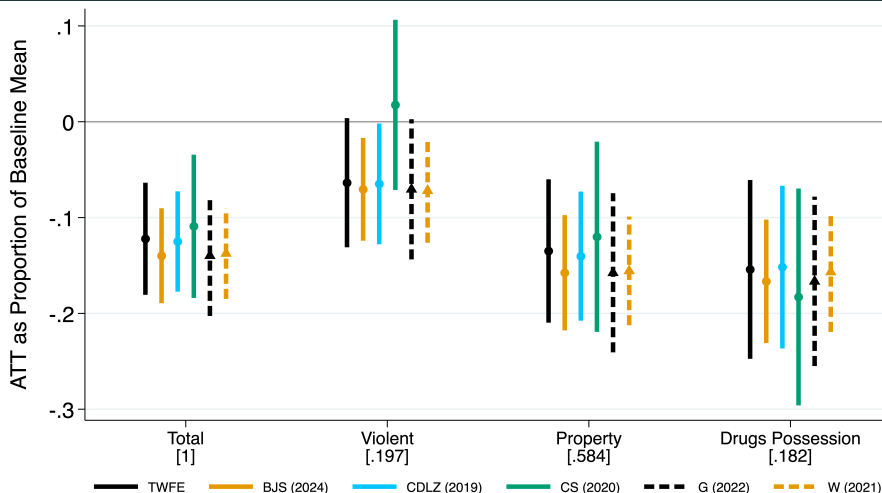
Characterizing the Location of Plasma Centers

[◀ data](#)
[reconcile ▶](#)

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	NIBRS-Based Plasma Centers			CODE-Based Plasma Centers			
	Locations with an Open Plasma Center	p-Value: NIBRS Plasma Location = Non-Plasma Location	p-Value: NIBRS Plasma Location = Non-NIBRS Plasma Location	Locations with an Open Plasma Center	p-Value: CODE Plasma Location = Non-Plasma Location	p-Value: CODE Plasma Location = Non-CODE Plasma Location	p-Value: CODE Plasma Location = NIBRS Plasma Location
Number of Plasma Centers	61			35			
(a) City-Level Location Characterization							
<i>Percentile Rank of:</i>							
Occupancy Rate	68.1 (20.6)	[0.000]	[0.009]	77.3 (19.2)	[0.000]	[0.000]	[0.022]
Ownership Rate	29.4 (23.8)	[0.000]	[0.072]	46.4 (35.7)	[0.466]	[0.000]	[0.013]
Unemployment Rate	50.1 (19.2)	[0.821]	[0.000]	43.2 (22.7)	[0.090]	[0.000]	[0.098]
Median HH Income	62.1 (21.2)	[0.000]	[0.017]	73.9 (17.4)	[0.000]	[0.000]	[0.004]
Poverty Rate	47.9 (24.2)	[0.519]	[0.001]	34.7 (23.8)	[0.000]	[0.000]	[0.007]
(b) Block Group-Level Location Characterization (Treated Cities Only)							
<i>Percentile Rank of:</i>							
Distance from City Center	24.7 (20.9)	[0.000]	[0.009]	47.2 (36.9)	[0.637]	[0.015]	[0.001]
Occupancy Rate	46.6 (24.8)	[0.288]	[0.330]	65.6 (23.3)	[0.000]	[0.000]	[0.000]
Ownership Rate	44.5 (27.9)	[0.124]	[0.241]	57.2 (28.7)	[0.166]	[0.001]	[0.039]
Unemployment Rate	47.1 (22.4)	[0.359]	[0.126]	41.3 (24.6)	[0.050]	[0.017]	[0.274]
Median HH Income	43.6 (25.6)	[0.051]	[0.165]	59.3 (22.9)	[0.024]	[0.000]	[0.001]
Poverty Rate	49.0 (27.7)	[0.820]	[0.035]	39.9 (26.7)	[0.034]	[0.000]	[0.095]

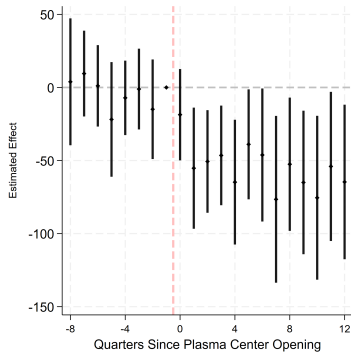
Alternative DD Estimators

emp strat

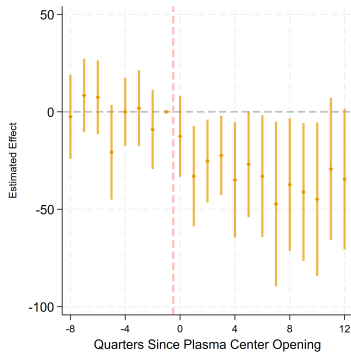


Notes: We present point estimates and 90% confidence intervals for our DD estimates from a variety of different estimators. These include: our baseline estimator [TWFE], and the estimators from [Borusyak et al. \[2024\]](#) [BJS (2024)], [Cengiz et al. \[2019\]](#) [CDLZ (2019)], [Callaway and Sant'Anna \[2021\]](#) [CS (2021)], [Gardner \[2022\]](#) [G (2022)], and [Wooldridge \[2021\]](#) [W (2021)]. In square brackets under each category label, we present the proportion of total crime accounted for by each crime category. Sample period: January 2006-December 2019. Data source: NIBRS.

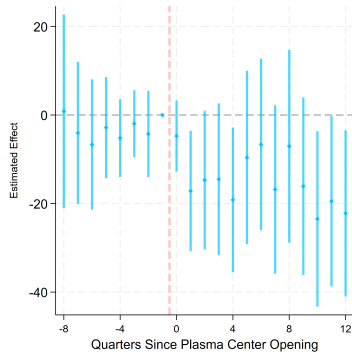
City-Level Event Study Estimates

[Emp Strategy](#)[DD](#)[add](#)[reconcile](#)[Long-run ES](#)

(a) Total Crime

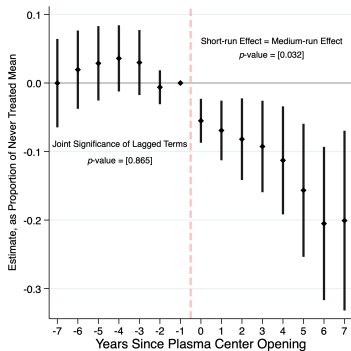


(b) Property Crime

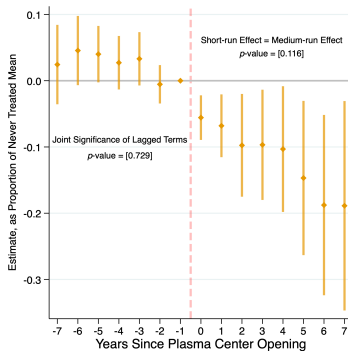


(c) Drug Possession

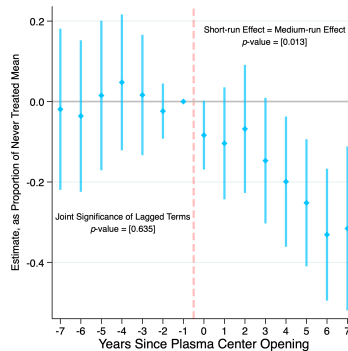
City-Level Event Study Estimates

[Emp Strategy](#)[DD](#)[add](#)[reconcile](#)[Short-run ES](#)

(a) Total Crime

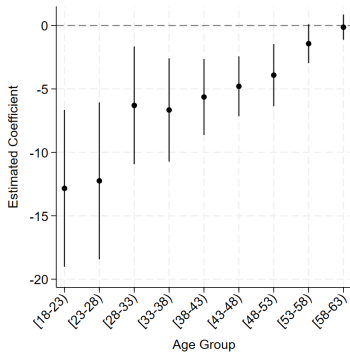


(b) Property Crime

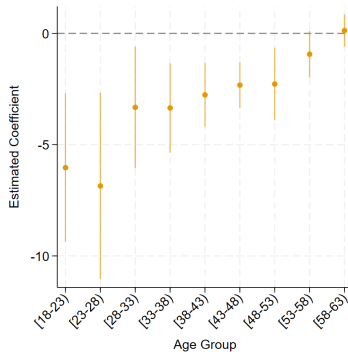


(c) Drug Possession

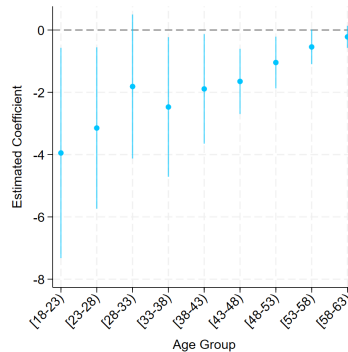
Effects by Age of Offender

[◀ back](#)

(a) Total Crime



(b) Property Crime

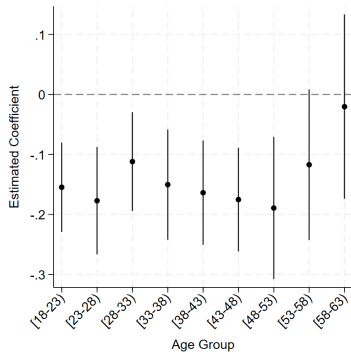


(c) Drug Possession

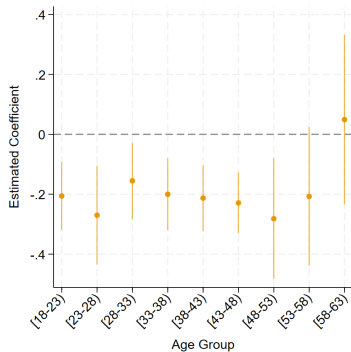
► Proportional effects

► Age reporting rates

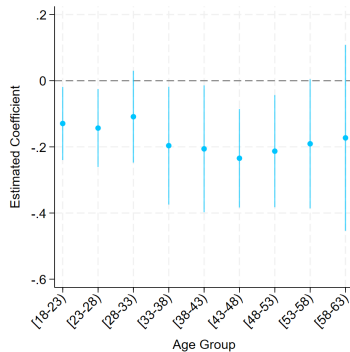
Proportional Effects by Age Relative to Untreated Average Crime Rate



(a) Total Crime



(b) Property Crime



(c) Drug Possession

Effects on Share of Offenders with Unreported Age

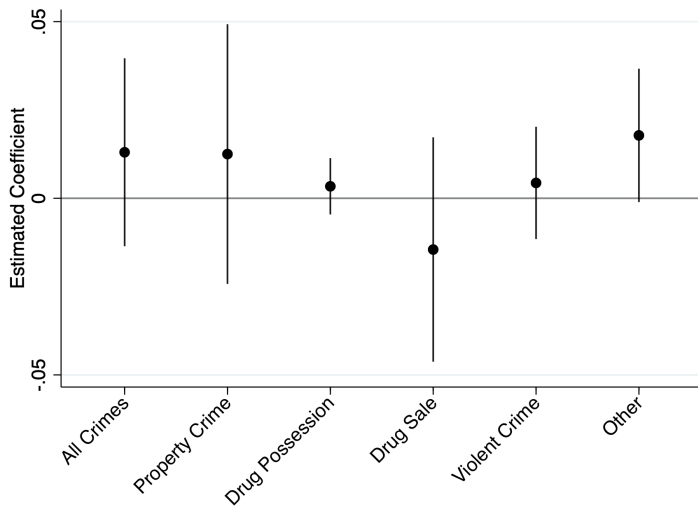
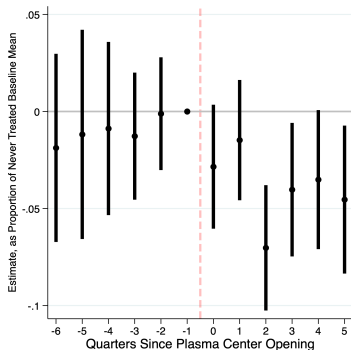
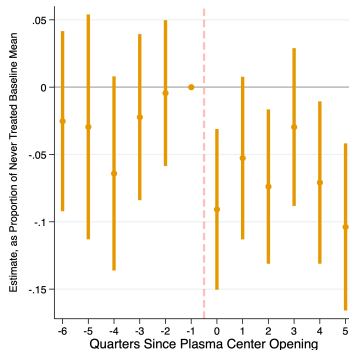
[◀ Back](#)

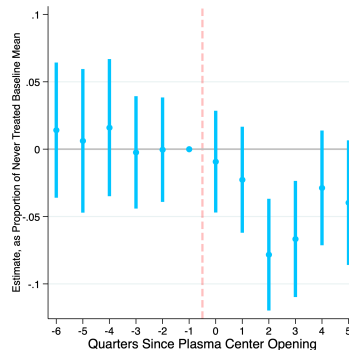
Figure 5: Event Study Estimates of Plasma Center Opening on Within-City Crime Outcomes



(a) Total Crime



(b) Violent Crime



(c) Property Crime

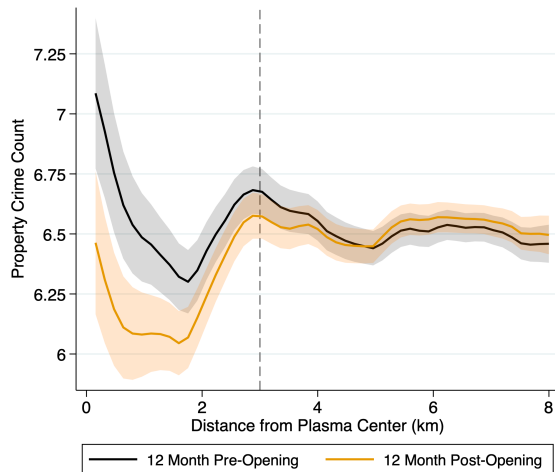
The Impact of the Opening of a Nearby Blood Plasma Center on Block-Group Crime Counts – PLACEBO

◀ Emp Strategy

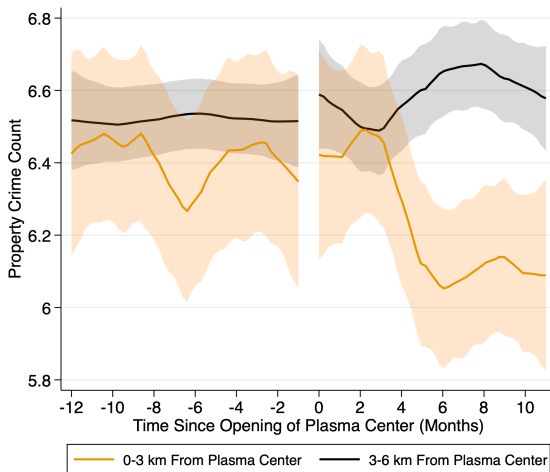
◀ DD

	(1)	(2)	(3)	(4)	(5)
	Total Crime	Violent Crime	Property Crime	Drug Crime	Other Crime
Near Plasma Center × Placebo Open	.167 (.201)	.0437 (.0596)	.0206 (.106)	.0468 (.0333)	.056 (.0626)
Proportion of Total Crime	[1.000]	[0.246]	[0.555]	[0.063]	[0.137]
$\bar{Y}_{NT,pre}$	11.5	2.83	6.24	.78	1.61
DD/ $\bar{Y}_{NT,pre}$	.0146 (.0175)	.0154 (.021)	.00331 (.017)	.06 (.0427)	.0348 (.0389)
Observations	64,440	64,440	64,440	64,440	64,440
Adjusted R^2	.935	.82	.891	.515	.923

Establishing a Distance-Based Cutoff



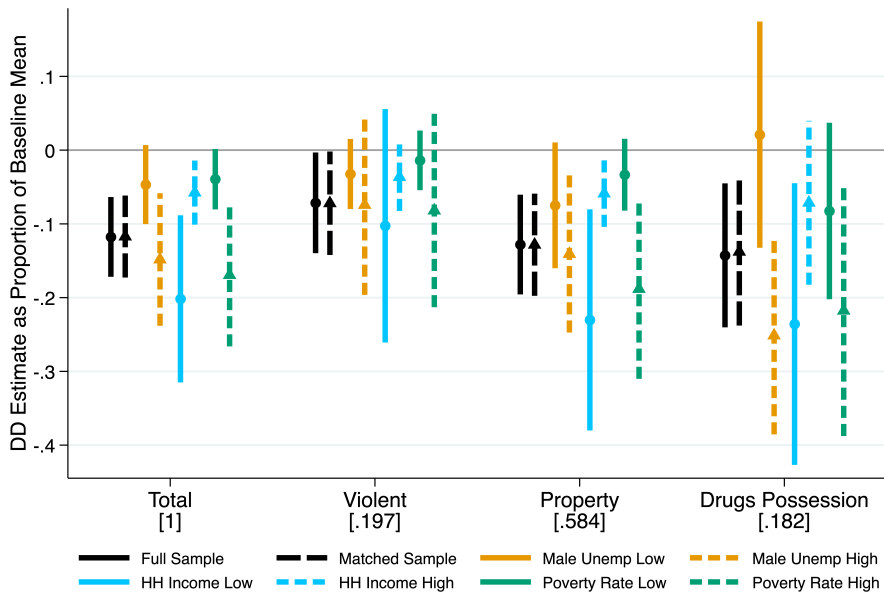
(a) The Crime-Distance Decay Curve



(b) Temporal Differences by Distance

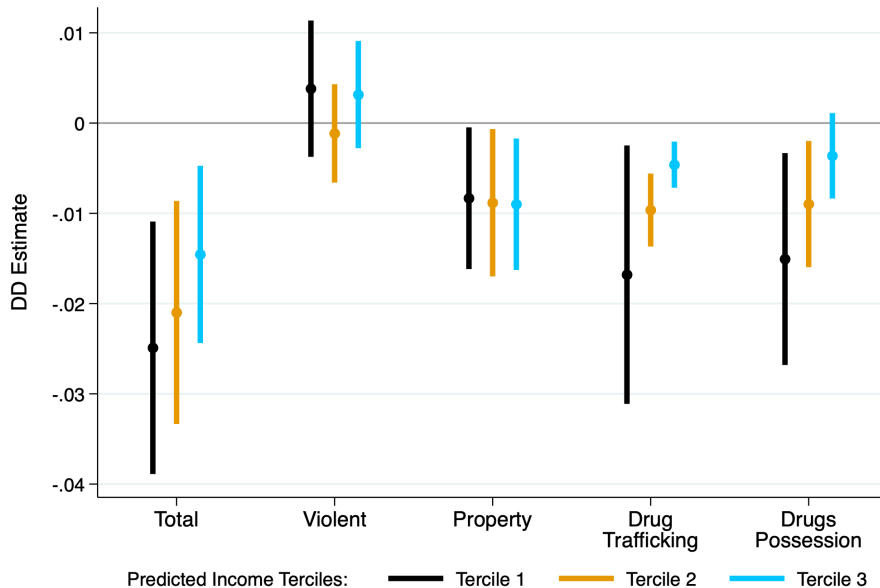
Notes: Local polynomial estimates and 90% confidence intervals are presented in both panels. We use Epanechnikov kernels, rule-of-thumb (ROT) bandwidths, and pilot bandwidths equal to 1.5 times the ROT bandwidths for standard error calculations. Data source: CODE.

Heterogeneity: Predetermined City Characteristics

[Mech](#)[recon](#)

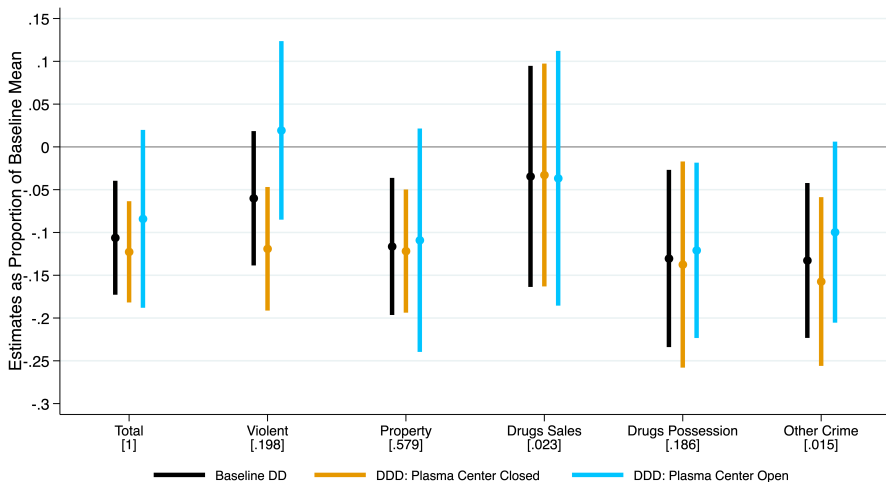
Heterogeneity: Predicted Individual Income

◀ Mech



DDD Analysis Using Clinic Opening Time Information

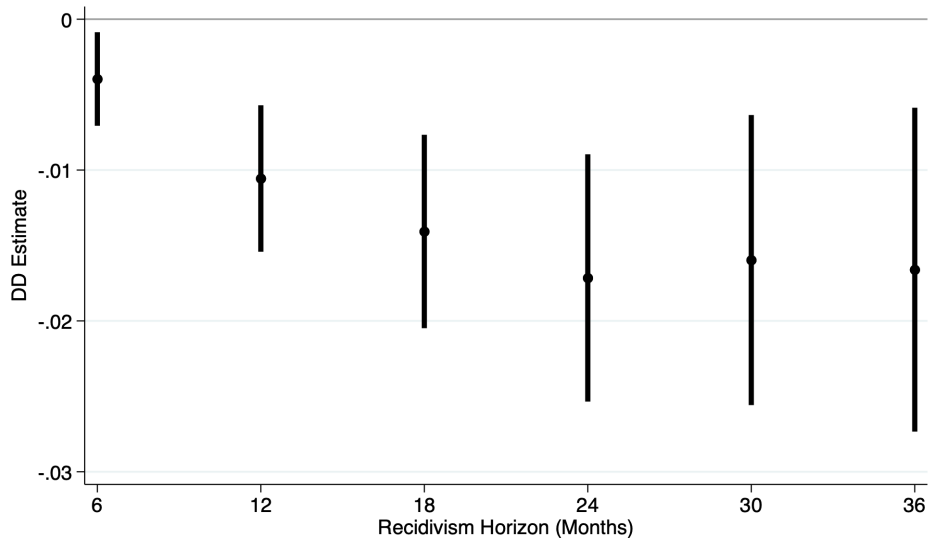
◀ Mechanisms



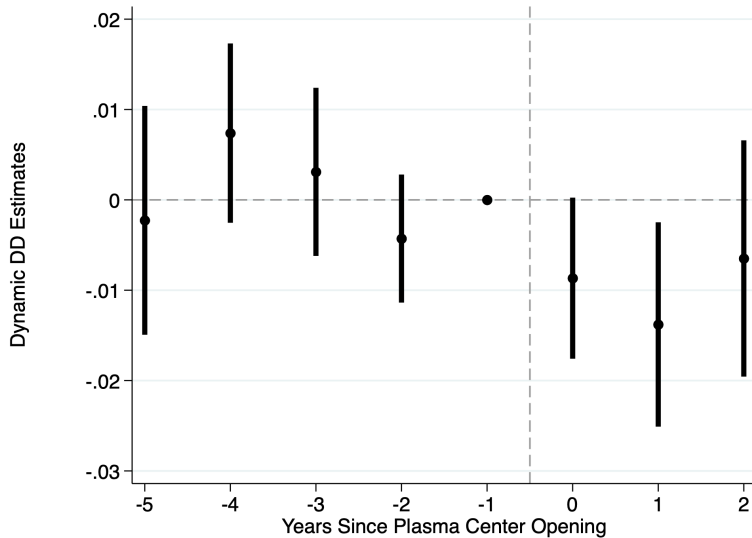
$$C_{cshdt} = \beta_1 D_{ct} + \beta_2 (D_{ct} \times open_{chdt}) + \gamma_c + \theta_{s \times t} + \tau_{c \times w \times q \times h} + \epsilon_{cshdt} ,$$

where h = hour; d = day of week; w = weekday/weekend; q = season/quarter

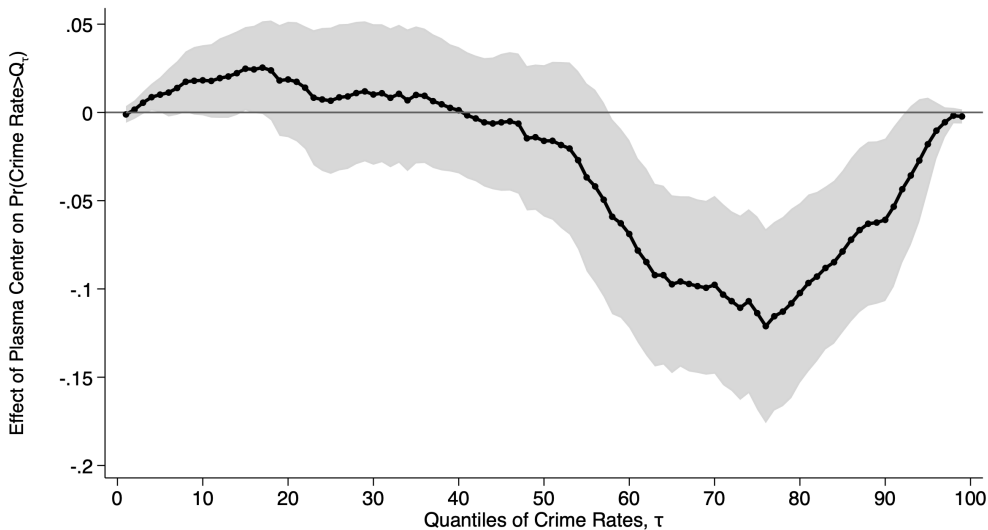
The Impact of Plasma Centers on Recidivism Rate (Raw Estimates)

[◀ Back](#)

Event Study: 3-year Recidivism

[◀ Back](#)

Distributional Effects of Plasma Centers on Crime

[◀ DD](#)[◀ add](#)

$$y_{cst} = \mathbb{1}[\text{crime}_{cst} > Q_\tau] \text{ for } \tau = [1, \dots, 99]$$

The Impact of Plasma Centers on City-Level Property Crime Rates

◀ DD

◀ add

	(1)	(2)	(3)	(4)
	Theft	Car Theft	Fraud	Other Property Crime
Plasma Center in City	-30.9*** (9.71)	-4.05** (1.63)	-4.33 (3.81)	-8.99* (5.19)
Proportion of Property Crime	[0.611]	[0.032]	[0.127]	[0.230]
$\bar{Y}_{NT}$	230	11.7	47.9	87.4
Plasma Center/ $\bar{Y}_{NT}$	-.134*** (.0421)	-.345** (.139)	-.0905 (.0796)	-.103* (.0595)
Number Treated Cities	53	53	53	53
Number Cities	1,185	1,185	1,185	1,185
Observations	198,744	198,744	198,744	198,744
Adjusted R^2	.665	.335	.425	.524

The Impact of Plasma Centers on City-Level Drugs Crime Rates

◀ DD

◀ add

	(1)	(2)	(3)	(4)	(5)	(6)
	Cocaine Possession	Heroin Possession	Marijuana Possession	Methamphetamine Possession	Other Possession	Possession, Drug Unknown
Plasma Center in City	-.6 (1.25)	-.724 (1.52)	-9.72** (4.5)	-4.67** (2.3)	.862 (1.66)	-2.19 (1.65)
Proportion of Drug Possession Crime	[0.053]	[0.039]	[0.546]	[0.100]	[0.106]	[0.156]
$\bar{Y}_{NT}$	6.2	4.66	65.4	11.9	12.8	18.4
Plasma Center/ $\bar{Y}_{NT}$	-.0968 (.202)	-.155 (.327)	-.149** (.0687)	-.392** (.193)	.0672 (.13)	-.119 (.09)
Number Treated Cities	53	53	53	53	53	53
Number Cities	1,185	1,185	1,185	1,185	1,185	1,185
Observations	198,744	198,744	198,744	198,744	198,744	198,744
Adjusted R^2	.166	.187	.408	.254	.338	.275

- We estimate larger crime decline using NIBRS than CODE.
 - Why?
1. Due to data limitations, CODE estimates are **temporally local** effects (12 month window).
NIBRS event study graphs: the impact of plasma centers on crime grows with time since opening
[\[Event study\]](#)
 2. CODE cities are relatively **positively selected** compared to NIBRS cities [\[plasma city Xs\]](#)
next slide: plasma centers matter more for crime in poorer cities
[\[DDD by income\]](#)
- Ideally, we would provide estimates using both empirical designs for overlap cities (in CODE and NIBRS)
 - **issue:** only 1 plasma center in both datasets